

The fecundity schedule: evidence, calibration precedents, and our fit

Fertility–housing project

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1. The object

In the sequential-fertility architecture, a household that attempts a birth succeeds within the model period with probability

$$\pi_j = \min\{1, \max\{0, 1 - \omega_1 e^{\omega_2(a_j-18)}\}\}, \quad \pi_j = 0 \text{ for } a_j \geq 45.$$

π_j is a primitive of the environment, not a calibrated SMM parameter: biology identifies it, and pinning it externally is what separates the biological margin from the choice margin — both π_j and the fertility logit scale κ_f smooth birth hazards, so a freely estimated π_j would leave κ_f unidentified. One sentence of economics: because π_j falls with age, postponing an attempt carries a real risk that the window closes, so the option value of waiting is priced and policy acquires a timing margin. Setting $\omega_1 = 0$ recovers certain conception — the production model — exactly.

2. What the demographic evidence says

Four sources anchor the shape; all figures below were verified against the sources cited (the 1986 table via a secondary source with explicit attribution, noted).

Conception probabilities by starting age. Léridon (2004) reports that under natural conditions 75% of women starting at age 30 conceive (with live birth) within one year, 66% at 35, and 44% at 40; *within four years* the corresponding figures are **91%, 84%, and 64%**. Because our model period is four years, these are direct empirical counterparts of π_j at ages 30, 35, and 40.

Infecundity by age. Menken, Trussell and Larsen (1986) report shares infertile rising from 7% at ages 20–24 to 15% (30–34), 22% (35–39) and 29% (40–44), with the chance of remaining childless when marrying in those bands rising from 6% to 64%.

Per-cycle fecundability. Dunson, Colombo and Baird (2002), from prospective data on 5,860 cycles, find per-cycle conception probabilities roughly twice as high at ages 19–26 as at 35–39, with decline beginning in the late twenties.

Natural sterility level. Baudin, de la Croix and Gobbi (2015) put natural sterility at 3.7% of couples, from Léridon’s (2008) use of the Henry database of 18th-century rural France (couples married at 20–24 with no effective fertility control).

3. How the literature calibrates it

Three strategies recur. (i) *Exponential fit to a permanent-infertility CDF*: Sommer (2016) specifies $p_t^I = \omega_1 e^{\omega_2 t}$ reaching one at 45 and fits it to the Trussell–Wilson (1985) point estimates of the

fraction permanently infertile by age (about 97% at 45); the fitted constants are shown graphically and not reported numerically in the paper. (ii) *Bounded logistic fit to conception-within- N -years*: de la Croix and Pommeret (2021) fit $\pi(t) = ae^{b-ct}/(d + e^{b-ct})$ to Léridon (2005, Table I) conception probabilities at seven durations, imposing $\pi(0) = 0.96$ from the 4% natural-sterility estimate; their reported fit is $(a, b, c, d) = (0.96, 5.53, 0.33, 0.012)$.¹ (iii) *A single external sterility share*: Baudin, de la Croix and Gobbi (2015) fix natural sterility at 3.7% and split it evenly by sex, leaving no age slope inside the fecund window.

Our approach is (ii) in spirit with (i)’s functional form: a two-parameter exponential in age, zero from 45, fit to within-window conception probabilities.

4. Our schedule and the fit

The four-year period makes the fit unusually direct: Léridon’s within-four-year probabilities *are* π at the corresponding ages. The planned procedure is least squares of π_j on the three Léridon points $\{0.91, 0.84, 0.64\}$ at ages 30/35/40, with the young-age level anchored by the natural-sterility estimate ($\pi \approx 0.96$ – 0.965 in the early twenties) and the terminal age at 45 (Sommer’s 97%-infertile benchmark; Menken et al.’s 64% childless for 40–44 marriages).

The schedule currently deployed in the E-series experiments, $(\omega_1, \omega_2) = (0.02, 0.134)$, was chosen by eye before this verification pass. Evaluated at the Léridon ages it gives:

	age 22	age 30	age 35	age 40
Current schedule π_j	0.966	0.899	0.805	0.618
Demographic target	0.960–0.963 ^a	0.91	0.84	0.64

^aone minus natural sterility (3.7–4%).

The deployed schedule sits on the evidence to within one to three percentage points at every anchor; the formal fit will therefore move the E-series results very little, and the “illustrative” caveat carried by the E1/E2 reports is, in substance, already discharged up to that refinement. The remaining formal step — running the least-squares fit, documenting it, and re-solving the E2 winner at the fitted schedule as a robustness line — is a half-day item and should precede any paper-facing use of the timing results.

Baudin, de la Croix, Gobbi (2015), *AER* 105(6).

de la Croix, Pommeret (2021), *JET* 193.

Dunson, Colombo, Baird (2002), *Human Reproduction* 17(5).

Léridon (2004), *Human Reproduction* 19(7).

Léridon (2005), *Rev. d’Épid. et de Santé Publique* 53.

Menken, Trussell, Larsen (1986), *Science* 233.

Sommer (2016), *JME* 83.

Trussell, Wilson (1985), *Population Studies* 39(2).

¹Two source cautions from our verification pass. The working paper’s displayed table prints $b = 3.5$ while the text reports the fitted $b = 5.53$ — an internal inconsistency we could not resolve against the paywalled published version. And the fit is sourced to Léridon (2005), a French-journal companion paper, not the more commonly cited Léridon (2004) *Human Reproduction* article.